

Bellabeat's users analysis

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Objective:

This analysis explores Fit bit user activity data to identify opportunities for Bellabeat to improve user engagement and health outcomes.

Installing and loading common packages and libraries

Creating dataframes from the dataset

```
daily_activity <- read.csv("dailyActivity_merged.csv")
sleep_day <- read.csv("sleepDay_merged.csv")
hourlyCalories <- read.csv("hourlyCalories_merged.csv")
```

Exploring a few key tables

```
head(daily_activity)
```

```
##           Id ActivityDate TotalSteps TotalDistance TrackerDistance
## 1 1503960366  4/12/2016      13162           8.50           8.50
## 2 1503960366  4/13/2016      10735           6.97           6.97
## 3 1503960366  4/14/2016      10460           6.74           6.74
## 4 1503960366  4/15/2016       9762           6.28           6.28
## 5 1503960366  4/16/2016      12669           8.16           8.16
## 6 1503960366  4/17/2016       9705           6.48           6.48
##   LoggedActivitiesDistance VeryActiveDistance ModeratelyActiveDistance
## 1                        0                1.88                   0.55
## 2                        0                1.57                   0.69
## 3                        0                2.44                   0.40
## 4                        0                2.14                   1.26
## 5                        0                2.71                   0.41
## 6                        0                3.19                   0.78
##   LightActiveDistance SedentaryActiveDistance VeryActiveMinutes
## 1                6.06                  0                25
## 2                4.71                  0                21
## 3                3.91                  0                30
## 4                2.83                  0                29
## 5                5.04                  0                36
## 6                2.51                  0                38
##   FairlyActiveMinutes LightlyActiveMinutes SedentaryMinutes Calories
## 1                 13                328                728    1985
## 2                 19                217                776    1797
## 3                 11                181               1218    1776
```

```
## 4          34          209          726          1745
## 5          10          221          773          1863
## 6          20          164          539          1728
```

```
colnames(daily_activity)
```

```
## [1] "Id"          "ActivityDate"
## [3] "TotalSteps"  "TotalDistance"
## [5] "TrackerDistance" "LoggedActivitiesDistance"
## [7] "VeryActiveDistance" "ModeratelyActiveDistance"
## [9] "LightActiveDistance" "SedentaryActiveDistance"
## [11] "VeryActiveMinutes" "FairlyActiveMinutes"
## [13] "LightlyActiveMinutes" "SedentaryMinutes"
## [15] "Calories"
```

```
head(sleep_day)
```

```
##           Id           SleepDay TotalSleepRecords TotalMinutesAsleep
## 1 1503960366 4/12/2016 12:00:00 AM                1                327
## 2 1503960366 4/13/2016 12:00:00 AM                2                384
## 3 1503960366 4/15/2016 12:00:00 AM                1                412
## 4 1503960366 4/16/2016 12:00:00 AM                2                340
## 5 1503960366 4/17/2016 12:00:00 AM                1                700
## 6 1503960366 4/19/2016 12:00:00 AM                1                304
## TotalTimeInBed
## 1          346
## 2          407
## 3          442
## 4          367
## 5          712
## 6          320
```

```
colnames(sleep_day)
```

```
## [1] "Id"          "SleepDay"          "TotalSleepRecords"
## [4] "TotalMinutesAsleep" "TotalTimeInBed"
```

```
head(hourlyCalories)
```

```
##           Id           ActivityHour Calories
## 1 1503960366 3/12/2016 12:00:00 AM        48
## 2 1503960366 3/12/2016 1:00:00 AM        48
## 3 1503960366 3/12/2016 2:00:00 AM        48
## 4 1503960366 3/12/2016 3:00:00 AM        48
## 5 1503960366 3/12/2016 4:00:00 AM        48
## 6 1503960366 3/12/2016 5:00:00 AM        48
```

```
colnames(hourlyCalories)
```

```
## [1] "Id"          "ActivityHour" "Calories"
```

All three dataset have the 'Id' field - this can be used to merge the dataset.

Understanding some summary statistics

How many unique participants are there in each dataframe? It looks like there may be more participants in the daily activity and hourly calories dataset than the sleep dataset.

```
n_distinct(daily_activity$Id)
```

```
## [1] 33
```

```
n_distinct(sleep_day$Id)
```

```
## [1] 24
```

```
n_distinct(hourlyCalories$Id)
```

```
## [1] 34
```

How many observations are there in each dataframe?

```
nrow(daily_activity)
```

```
## [1] 940
```

```
nrow(sleep_day)
```

```
## [1] 413
```

```
nrow(hourlyCalories)
```

```
## [1] 24084
```

For the daily activity dataframe:

```
daily_activity %>%  
  select(TotalSteps,  
         TotalDistance,  
         SedentaryMinutes,  
         VeryActiveMinutes,  
         LightlyActiveMinutes,  
         FairlyActiveMinutes) %>%  
  summary()
```

```
##      TotalSteps      TotalDistance      SedentaryMinutes      VeryActiveMinutes  
## Min.   :    0      Min.   : 0.000      Min.   :    0.0      Min.   :    0.00  
## 1st Qu.: 3790      1st Qu.: 2.620      1st Qu.: 729.8      1st Qu.:    0.00  
## Median : 7406      Median : 5.245      Median :1057.5      Median :    4.00  
## Mean   : 7638      Mean   : 5.490      Mean   : 991.2      Mean   :   21.16  
## 3rd Qu.:10727      3rd Qu.: 7.713      3rd Qu.:1229.5      3rd Qu.:   32.00  
## Max.   :36019      Max.   :28.030      Max.   :1440.0      Max.   :  210.00  
## LightlyActiveMinutes FairlyActiveMinutes  
## Min.   :    0.0      Min.   :    0.00  
## 1st Qu.:127.0      1st Qu.:    0.00  
## Median :199.0      Median :    6.00  
## Mean   :192.8      Mean   :   13.56  
## 3rd Qu.:264.0      3rd Qu.:   19.00  
## Max.   :518.0      Max.   :  143.00
```

On the average a user is very active for 21mins, lightly active for 193 mins, fairly active for about 14 mins, and sedentary for 991mins in a day.

For the sleep dataframe:

```
sleep_day %>%  
  select(TotalSleepRecords,  
         TotalMinutesAsleep,
```

```
TotalTimeInBed) %>%
summary()
```

```
## TotalSleepRecords TotalMinutesAsleep TotalTimeInBed
## Min. :1.000 Min. : 58.0 Min. : 61.0
## 1st Qu.:1.000 1st Qu.:361.0 1st Qu.:403.0
## Median :1.000 Median :433.0 Median :463.0
## Mean :1.119 Mean :419.5 Mean :458.6
## 3rd Qu.:1.000 3rd Qu.:490.0 3rd Qu.:526.0
## Max. :3.000 Max. :796.0 Max. :961.0
```

This shows that on the average, users get about 7 hours of sleep even though they spend over 7 hours in bed.

For the hourly calories:

```
library(dplyr)
library(lubridate)
ActivityHour <- read.csv("hourlyCalories_merged.csv")
ActivityHour <- ActivityHour %>%
  mutate(
    ActivityHour = mdy_hms(ActivityHour), # convert text to datetime
    Date = as.Date(ActivityHour), # just the date
    Hour = hour(ActivityHour), # 0-23
    Weekday = wday(ActivityHour, label = TRUE) # Mon, Tue...
  )
```

```
hourlyCalories %>%
select(ActivityHour,
  Calories,
) %>%
summary()
```

```
## ActivityHour      Calories
## Length:24084      Min.   : 42.00
## Class :character  1st Qu.: 61.00
## Mode  :character  Median : 77.00
##                      Mean   : 94.27
##                      3rd Qu.:104.00
##                      Max.   :933.00
```

On the average, customers burn 94 calories per hour.

plotting a few explorations

- Insight 1: time in bed vs time asleep

```
library(dplyr)
library(ggplot2)

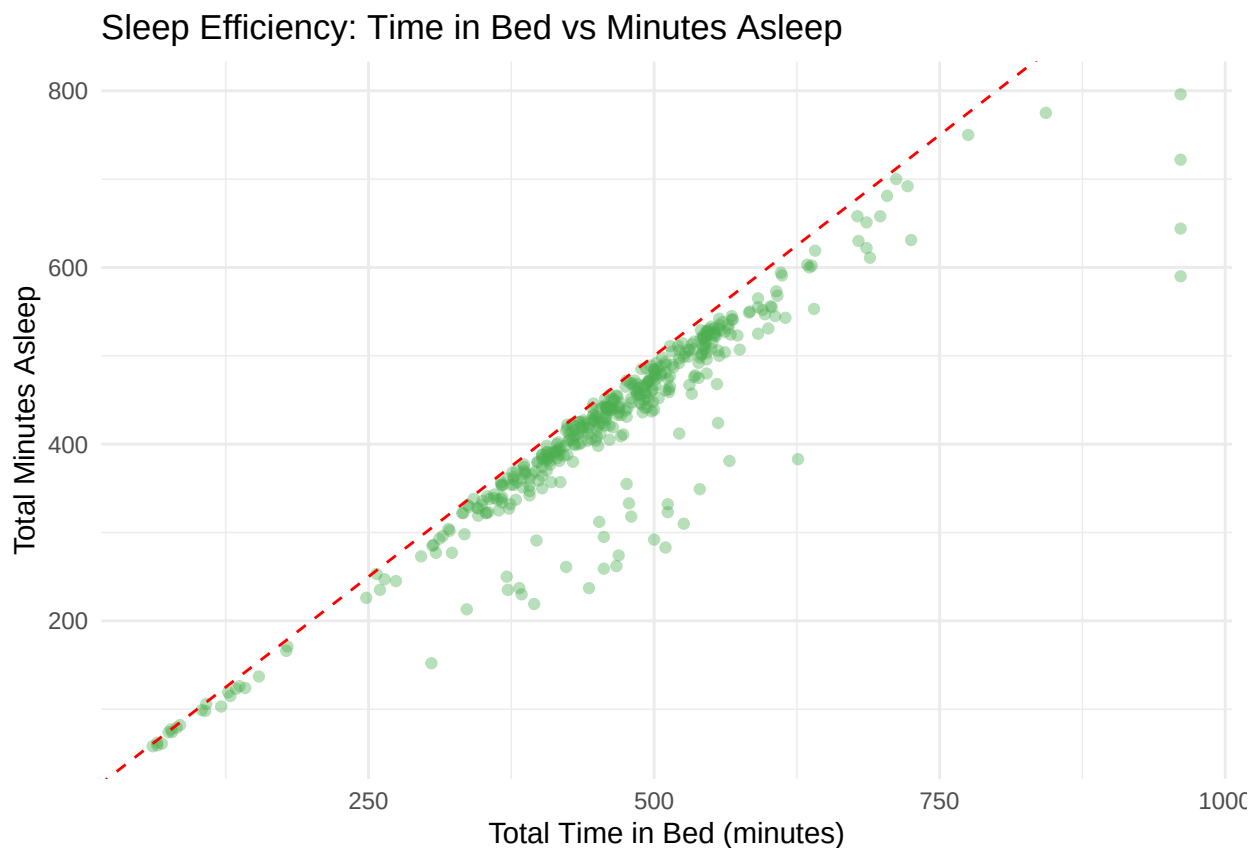
# 1. Load the sleep data
SleepDay <- read.csv("sleepDay_merged.csv")

# 2. Create a new column: MinutesAwakeInBed
SleepDay <- SleepDay %>%
  mutate(MinutesAwakeInBed = TotalTimeInBed - TotalMinutesAsleep)
```

```
# 3. Check it
head(SleepDay)
```

```
##           Id           SleepDay TotalSleepRecords TotalMinutesAsleep
## 1 1503960366 4/12/2016 12:00:00 AM                1                327
## 2 1503960366 4/13/2016 12:00:00 AM                2                384
## 3 1503960366 4/15/2016 12:00:00 AM                1                412
## 4 1503960366 4/16/2016 12:00:00 AM                2                340
## 5 1503960366 4/17/2016 12:00:00 AM                1                700
## 6 1503960366 4/19/2016 12:00:00 AM                1                304
## TotalTimeInBed MinutesAwakeInBed
## 1              346              19
## 2              407              23
## 3              442              30
## 4              367              27
## 5              712              12
## 6              320              16
```

```
# 4. PLOT 1: Minutes Asleep vs Time In Bed
ggplot(SleepDay, aes(x = TotalTimeInBed, y = TotalMinutesAsleep)) +
  geom_point(alpha = 0.4, color = "#4CAF50") +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "red") + # perfect sleep line
  labs(title = "Sleep Efficiency: Time in Bed vs Minutes Asleep",
        x = "Total Time in Bed (minutes)",
        y = "Total Minutes Asleep") +
  theme_minimal()
```



key finding: Dashed red line = perfect sleep, so while the majority of Bellabeat users demonstrate high sleep efficiency, a segment spends significant time awake in bed.

Insight 2: Users by Activity level

```
library(dplyr)
library(ggplot2)
library(scales) # for % labels

##
## Attaching package: 'scales'

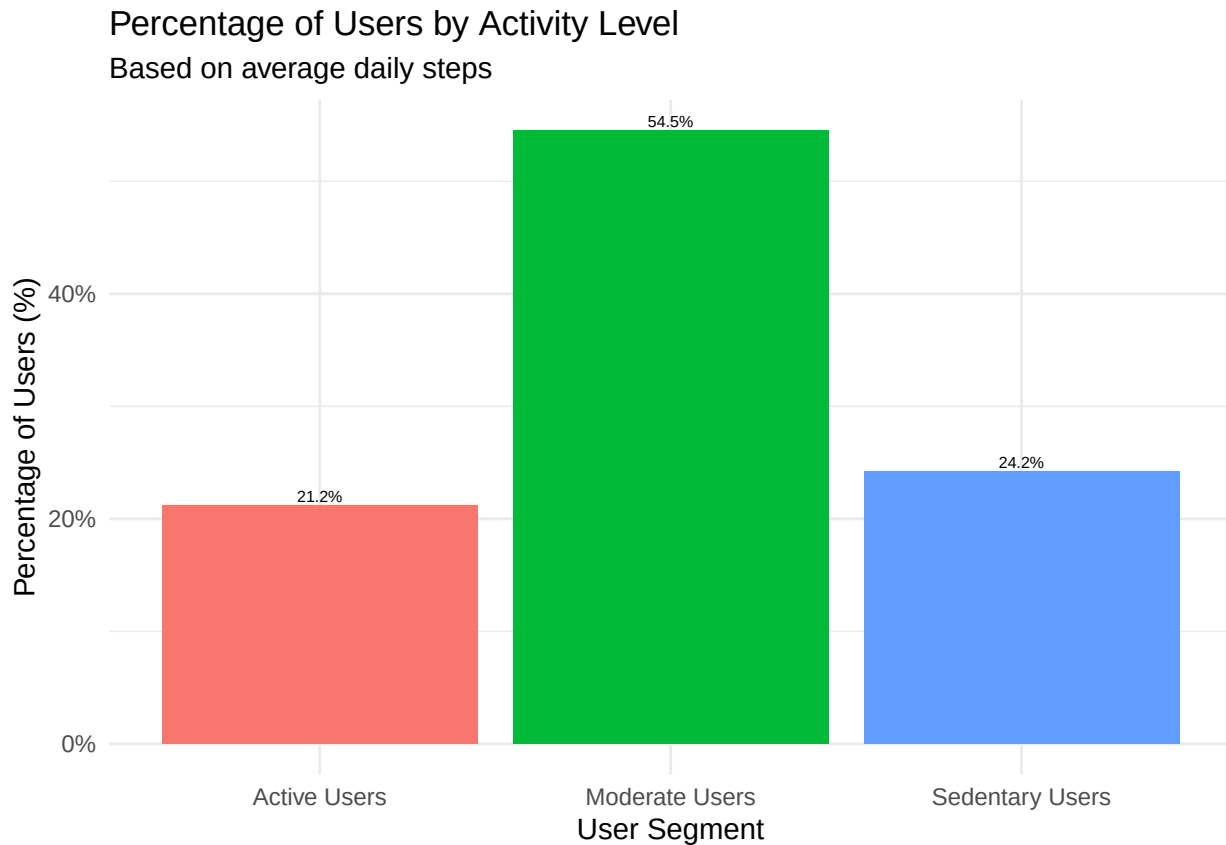
## The following object is masked from 'package:purrr':
##
##      discard

## The following object is masked from 'package:readr':
##
##      col_factor

# 1. Create user segments
user_segments <- daily_activity %>%
  group_by(Id) %>%
  summarize(AvgSteps = mean(TotalSteps, na.rm = TRUE), .groups = 'drop') %>%
  mutate(UserSegment = case_when(
    AvgSteps < 5000 ~ "Sedentary Users",
    AvgSteps < 10000 ~ "Moderate Users",
    TRUE ~ "Active Users"
  ))

# 2. Count + Calculate Percentage
segment_counts <- user_segments %>%
  count(UserSegment) %>%
  mutate(Percent = n / sum(n) * 100) # <-- this makes the %

# 3. Plot with % on top of bars
ggplot(segment_counts, aes(x = UserSegment, y = Percent, fill = UserSegment)) +
  geom_col() +
  geom_text(aes(label = paste0(round(Percent, 1), "%")), vjust = -0.3, size = 2.0) + # add % labels
  labs(
    title = "Percentage of Users by Activity Level",
    subtitle = "Based on average daily steps",
    x = "User Segment",
    y = "Percentage of Users (%)"
  ) +
  scale_y_continuous(labels = percent_format(scale = 1)) + # show y-axis as %
  theme_minimal() +
  theme(legend.position = "none")
```



key finding: Bellabeat has more than half moderately active users, sedentary users at 24.2% and very active users at 21.2% only.

Insight 3: steps versus calories

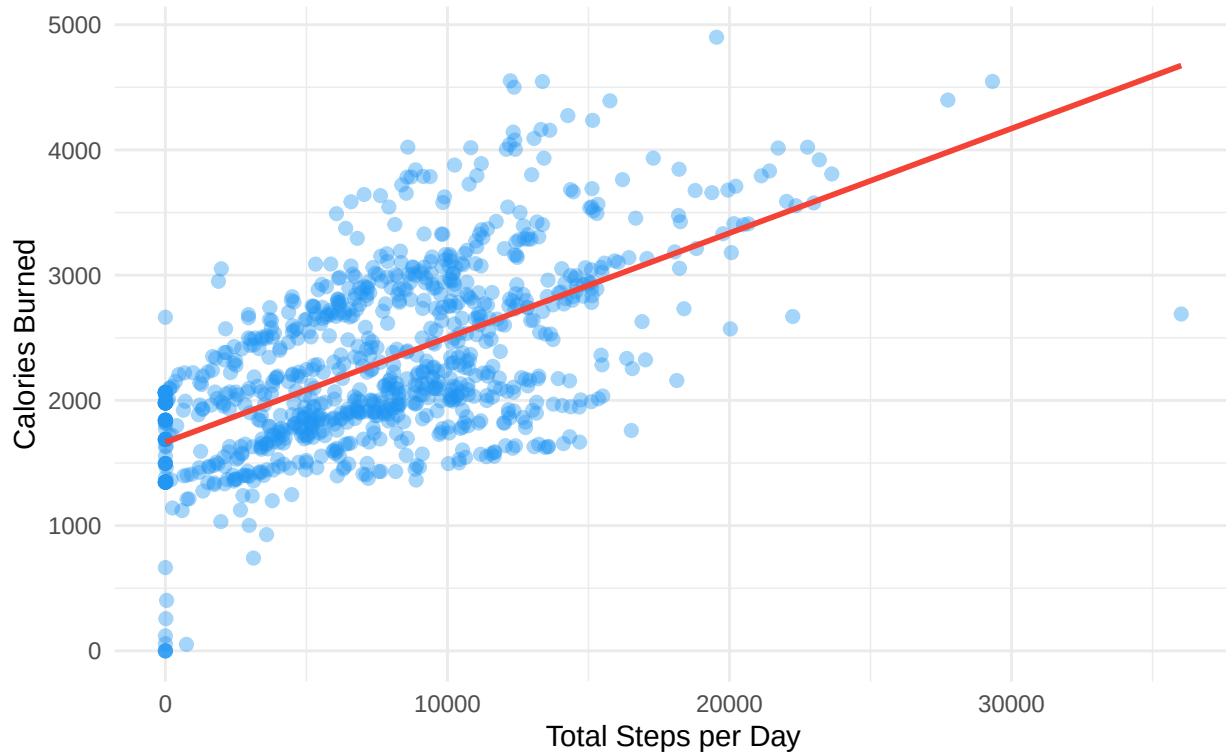
```
library(ggplot2)
library(dplyr)

ggplot(daily_activity, aes(x = TotalSteps, y = Calories)) +
  geom_point(alpha = 0.4, color = "#2196F3", size = 2) +
  geom_smooth(method = "lm", color = "#F44336", se = FALSE) + # adds trend line
  labs(
    title = "Relationship Between Steps and Calories Burned",
    subtitle = "More steps generally leads to more calories burned",
    x = "Total Steps per Day",
    y = "Calories Burned"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Relationship Between Steps and Calories Burned

More steps generally leads to more calories burned



key finding: There is a positive correlation. As Total steps per day goes up, Calories Burned also goes up as shown in the red line.

Insight 4: Steps by day of the week.

Chunk A: Clean the data

```
library(lubridate)
library(dplyr)
```

```
daily_activity <- read.csv("dailyActivity_merged.csv") %>%
  mutate(
    ActivityDate = mdy(ActivityDate),
    Weekday = weekdays(ActivityDate)
  )
```

Chunk B: Summarize

```
avg_steps_by_weekday <- daily_activity %>%
  group_by(Weekday) %>%
  summarize(AvgSteps = mean(TotalSteps, na.rm = TRUE), .groups = 'drop') %>%
  mutate(Weekday = factor(Weekday, levels = c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday")))
```

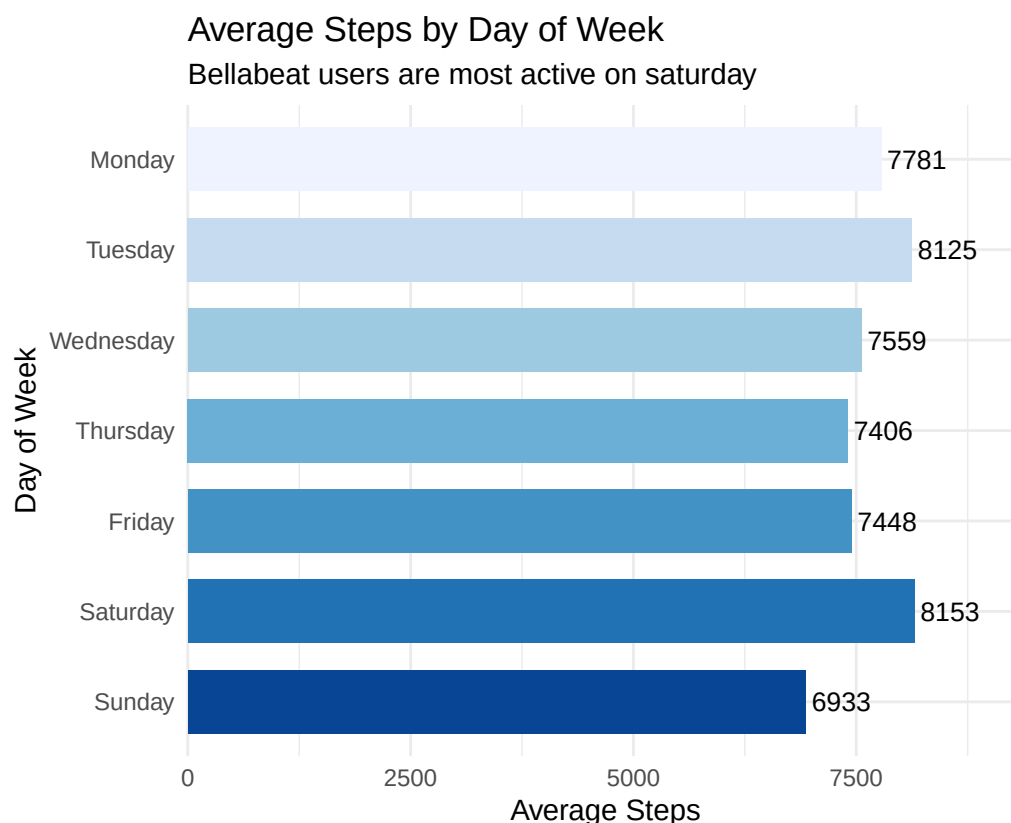
```
library(dplyr)
library(ggplot2)
```

```
ggplot(avg_steps_by_weekday, aes(x = Weekday, y = AvgSteps, fill = Weekday)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = round(AvgSteps, 0)), hjust = -0.1, size = 3.5) +
```

```

labs(
  title = "Average Steps by Day of Week",
  subtitle = "Bellabeat users are most active on saturday",
  x = "Day of Week",
  y = "Average Steps"
) +
scale_x_discrete(limits = rev(levels(avg_steps_by_weekday$Weekday))) +
scale_y_continuous(expand = expansion(mult = c(0, .15))) +
scale_fill_brewer(palette = "Blues") +
coord_flip(clip = "off") +
theme_minimal() +
theme(
  legend.position = "none",
  plot.margin = margin(10, 80, 10, 10)
)

```



key finding: Bellabeat users are most active on Saturday with average of 8153 steps, there is a drop in weekday activities from Wednesday to Thursday and the least active is Sunday with 6933 steps.

Recommendations:

1. Bellabeat should personalize goals using this data. Feature idea: 'Walk 2,000 more steps to burn 100 calories today'. This turns the correlation into motivation and helps users see direct ROI on movement.
2. Bellabeat should focus on 2 features: 1. 'Push to 10k' challenges to convert moderate users to Active, and 2. 'Gentle start' programs with hourly nudges for sedentary users.

3. Bellabeat can add features like sleep quality scoring tracking to help users improve rest, not just duration.
4. Bellabeat should launch a ‘Wednesday Reset’ challenge to boost engagement during the week and ‘Sunday finish strong’ reminder to encourage users to maintain Saturday energy.

Conclusion:

By implementing these data-driven features, Bellabeat can help users build more consistent habits, increase engagement, and improve overall health outcomes. This positions Bellabeat to better compete by focusing on weekly behavior patterns and reducing sedentary lifestyles.